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Urban Road Surface Analysis via Dashcam: Benchmarking YOLOv8 and YOLOv11 for Pothole Detection

Kimmi Gupta[a,b,c]*, Ch. Ravi Sekhar[a,b], Dileep Kumar Yadav[c], Lahar Srivastava[c]

[a]*Transportation Planning and Environment Division, CSIR-Central Road Research Institute, New Delhi, Delhi -110025, India*
[b]*Academy of Scientific and Innovative Research(AcSIR), Ghaziabad -201002, India*
[c]*School of Computer Science Engineering and Technology, Bennett University, Greater Noida – 201310, India*

Abstract

Potholes on Urban Roads are increasingly becoming a major global concern, posing serious risks to road users, vehicles on the road, and traffic safety, contributing to the probability of road accidents. Timely and correct detection of these potholes plays a pivotal role in reducing the risk of accidents, increasing the service life of the roads, ensuring pedestrian safety, and improving overall transportation efficiency. However, automatic pothole detection still poses difficulties due to aspects such as low image contrast, texture similarity, shadow artifacts, and intensity inhomogeneity. The primary objective of this research is to develop a model that detects potholes in real-time by enhancing the YOLOv8-x and YOLOv11-x and extract features such as classification of vehicles, information board, and pedestrian identification. The proposed work improves the performance by applying pre-processing to the input video which are framewise annotated using the roboflow. The model further uses saturation method to enhance the detection and auto-orient method to prevent frame loss. This research evaluates two non-YOLO models: SSD-MobileNet and Faster R-CNN on the same dataset using identical training and testing splits, then compared with considered YOLO models. The proposed results are significantly outperformed SSD-MobileNet and Faster R-CNN.

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Keywords: YOLOv8; YOLOv11; Deep Learning; Potholes; Auto-orient Method

* Corresponding author. Tel.: +917300948976
E-mail address: kimmi2391@gmail.com

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1. Introduction

Road safety is a critical factor in highway design, with the health and well-being of all users being the top priority for everyone. During the planning of a highway that preserves exquisite and environmental standards, it is equally essential to ensure that the highway operates efficiently, safely, and economically sustainable. The primary challenges are intra-class similarity (confusion with shadows, cracks, or patches) and inter-class variability (pothole shape/size), which result in false positives. Additionally, poor illumination, water pooling, and motion blur during vehicle capture are examples of unfavourable environmental variables that significantly reduce image quality. More than 385 million vehicles have been registered in India at this point, and the number is rapidly increasing [1]. India has been dependent on automobiles for a very long time. Road construction is now being carried out at a rapid pace by the government of India. However, road maintenance is made more difficult by the presence of large vehicles and poor drainage. Lack of proper maintenance of highways leads to the formation of potholes, which in turn leads to accidents involving vehicles. The existence of potholes in the roadway can put the life of a person who is driving in danger. Between 2018 and 2022, potholes alone were responsible for 8963 fatalities [2]. Potholes were identified as the root cause of 1% of the roughly 60,000 instances that occurred on Indian roads in the 2020 Road Accidents in India Report [3]. In addition to damaging vehicles, potholes affect pedestrians and drivers as well. Numerous techniques, including the use of sensors and specialized gear mounted on cars, have been developed to detect potholes. Although machine learning (ML)-based detection is better than earlier methods, developing feature extractors frequently requires a lot of effort and knowledge, and accuracy may still be increased. However, automatic pothole-detecting technologies are becoming more and more common due to enhanced in computer vision and deep learning algorithms. The most promising deep learning strategy for recognizing items, including potholes, is the YOLO method. YOLO is a kind of neural network that recognizes things in live video feeds by using object detection and classification methods. Its great speed and precision in object detection have led to its enormous popularity. The main key points of our research are as follows and described in Fig.1.

- In this paper, to detect the potholes and other classes such as vehicle identification, Information board and Person/pedestrian identification, YOLOv8-x, and YOLOv11-x models are used and compared to measure the precision, recall and F_1-score.
- Real-time dataset is collected using dashcam mounted on car for Inter-Urban roads that detects potholes and other classes, and this dataset in the form of video is then converted into frames. These frames are then pre-processed with the Auto-oriented, resized into 640 by 640, and Saturated -25% to +25%.
- The Proposed Method is compared with the existing method and performance is measured in terms of precision, recall and F_1-score on different confidence scores.

The major components of research are as follows: Section II reviews the related work on pothole detection; Section III presents the proposed methodology; Section IV discusses the experimental results and Section V concludes the Paper.

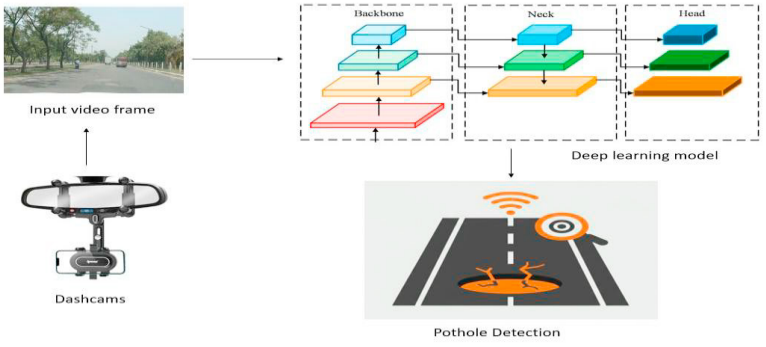


Fig. 1: Process of Proposed Pothole Detection

2. Related Work

India's vast and ageing road network plays a vital role in the nation's economy, connectivity, and public safety. However, the current methods for inspecting road conditions largely rely on manual field surveys, which are not only laborious and slow but also prone to human bias and subjectivity. The conventional technique cannot be scaled up or maintained in a country like India, where thousands of kilometres of roads need to be inspected and maintained on a regular basis. If left unaddressed, the damage to a road's surface, including potholes and cracks, can cause the road to deteriorate at a rapid pace, raise the costs associated with running a vehicle, and even cause accidents that could result in serious injury or death. With the growth of computer vision and the rise of AI and deep learning, particularly in the field of computer vision, there exists an unprecedented opportunity to revolutionise road health monitoring. It is possible for us to guarantee evaluations that are quicker, more dependable, and more cost-effective by automating the processes of detection, segmentation, and analysis of pavement distress. Deep learning models such as YOLOv8, YOLOv11[21] for real-time pavement analysis enable infrastructure agencies to monitor roads more efficiently, and it also creates opportunities for the development of more intelligent cities, predictive maintenance systems, and more secure public transportation. In the end, the goal of our endeavour is to make a significant contribution to the sustainability of national infrastructure and the welfare of the public by providing solutions that are intelligent, scalable, and actionable. Although ML-based detection is an advance over previous techniques, creating a feature extractor often needs significant work and expertise, and accuracy may still be improved [10],[11]. However, with advancements in computer vision and deep learning algorithms, automated pothole-detecting technologyg are gaining popularity [4], [5]. The You Only Look Once (YOLO) method is the most promising DL technique for identifying objects, including pothole identification applications. YOLO is a type of neural network that uses object detection and classification techniques. Its incredible speed and precision in object detection contributed to its tremendous popularity. However, several approaches have been explored before, but they all have significant limitations, such as slower result production and less stable implementations [8],[9]. Roman-Garay et al. [12] proposed a pothole identification and assessment system that integrates 2D image data with 3D point clouds obtained with an Intel RealSense D435i camera. They extracted depth and diameter measurements from the point clouds after using a Segformer deep learning model to partition potholes semantically and implemented a fuzzy logic algorithm to assess each pothole's level of severity. Hoseini et al. [13] presented an innovative method to identify and monitor potholes by installing standard cameras on cars and their system tracks weather conditions from different places using YOLOv5. Bhavana et al. [1] Proposed a deep learning model that works on real-time pothole detection using YOLOv8.Their approach first uses edge detection and applies a specific filter (CAGF) to video frames to decrease noise and then their approach uses YOLOv8 for different kinds of Potholes likes cracks, patches, and oil stains. Dhingra et al. [14] applies various machine learning models: YOLO, SSD, CNN, and HOG with SVM for potholes detection. They also apply erosion and dilation for the estimation of pothole size. Sathvik et al. [15] proposed a method for pothole detection using CNN and YOLOv7. They evaluate road conditions and find potholes using the GPS and uses a smartphone camera mounted on dashboard making it a better way to use. Al-Shaghouri et al. [16] developed a pothole detection system in real time that uses deep learning models like SSD, YOLOv3, and YOLOv4 Their method proved They have used dashcams and internet sources to collect data and shows that at 20 FPS, YOLOv4 performed better than other models and their method is appropriate for real-world road safety and autonomous car applications as it could detect potholes up to 100 meters in advance. Fortin et.al. [18] demonstrated significant improvements for YOLOv11-nano model in real-time performance using architectural optimizations such as neural architecture search (NAS) and depth wise separable convolution that reduce computational load and size of the model. It also preserves acceptable levels of detection accuracy.

3. Methodology

This section discusses the implementation of the YOLOv11-x and YOLOv8-x models in real-time datasets and compares the proposed model with existing variants of the YOLO models. Fig.2 depicts the architecture of the proposed model.

3.1 Datasets

In this section, the model uses real-time datasets and compares the model with different variants of YOLO models. The dashcam is used to collect real-time video, and this video is then processed to extract the frames. The dataset includes classes such as potholes, cars, bikes, trucks, mini-trucks, information boards, and pedestrian. The dataset is publicly available on Roboflow, which includes 2266 frames from which 1977 frames are kept for training the model, 141 for validating the model, and 148 for testing the model.

Table 1: Hyperparameters used in the Proposed model

Hyperparameter	Values
Optimizer	Adam
Learning Rate	0.01
Momentum	0.937
Weight Decay	0.0005
Image Size	640 × 640

In this paper, the collected dataset goes through the end-to-end pre-processing pipeline to maintain reliability, data consistency, and quality before training the AI models. The raw dataset consists of 2266 video frames collected from multiple sources, such as dashcam recordings of Inter-urban roads, traffic authorities and CCTV Surveillance cameras. As the data comes from different sources, there exist inconsistencies, gaps, and repetition of data that need to be handled before giving the data input to the machine learning algorithms. The hyperparameter used the batch size of 16 and the number of epochs with train 100, and the learning rate is set as 0.0001, and the augmentation besides saturation is used as resize and scaling. The dataset can be accessed using the link in the reference [17].

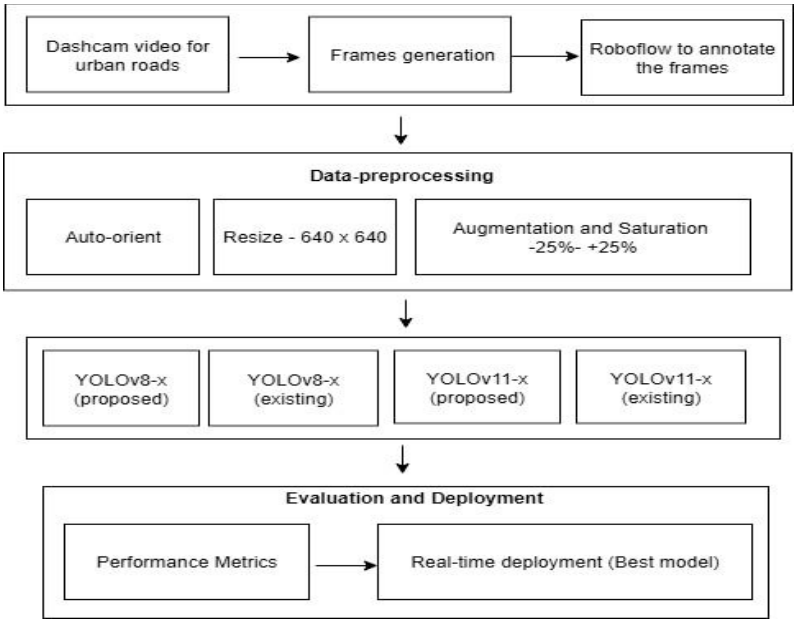


Fig. 2: The working of the Proposed model

3.2 Auto-orientation process

In the pre-processing, the auto-orient process is applied to the input data, which is used for automatic correction of the orientation of the frames based on metadata. The dataset captured by the dashcam device, where the frames appear upright on the screen, is stored in a rotated or flipped format to standardize the orientation of all frames for consistent training. In the next step, the orientation corresponds to a transformation matrix applied to the image coordinate

system. Considering the image to be defined as a 2D matrix $I(x,y)$, where $x \in [0, W-1]$ and $y \in [0, H-1]$.
Let the transformations be defined as follows:

Horizontal Flip: $$x' = W - 1 - x \quad (1)$$

Vertical Flip: $$y' = H - 1 - y \quad (2)$$

Rotation 90° Clockwise:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} H-1 \\ 0 \end{bmatrix} \quad (3)$$

Rotation 180°: $$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} W-1 \\ H-1 \end{bmatrix} \quad (4)$$

Rotation 270° Clockwise (or 90° Counter-Clockwise):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ W-1 \end{bmatrix} \quad (5)$$

3.3 Resize Process

The resizing method converts the input video into a frame of 640 x 640 pixels, the initial stage of resampling the frame to a 2D discrete signal to a new coordinate grid. Then it maps the new pixel locations to the original coordinate space. The interpolation pixel values are used to estimate intensity at new locations and transform an image from original size W×H to a new fixed size (here, 640×640) using resampling and interpolation.

3.4 Saturation Method

The Saturation method is used to adjust the hyperparameter used during the training process in the data augmentation phase. It also involves randomly changing the color intensity of the input images to +25% to -25% using the factor parameter (Hue, Saturation, Value). It also helps to improve the model's robustness to learn independently feature of specific color vividness.

3.5 Overall Architecture

With the development of a unified deep neural network technique, the YOLO framework improved object detection by performing both bounding box prediction and regression, as well as object classification in a single pass. The method integrated a design that represented a major shift in changes from traditional two-stage detection pipelines, enabling an end-to-end training process through a fully differentiable structure. The YOLO model is composed of three major key components such as backbone, neck, and head. A convolutional neural network layer is used within the backbone, which serves as the main feature extractor, to transform the processed image inputs into multi-scale feature maps. As an intermediary module that employs specialised layers to acquire and refine features across various scales, the neck enhances the backbone layer. Furthermore, the head module is responsible for using the processed feature maps to generate output predictions, which include object localization and class labels. The YOLOv11 model is an upgrade over YOLOv8 since it uses architecture refinement and parameter optimization to provide better object detection. Through the development of a unified deep neural network technique, the YOLO framework revolutionised object detection. Fig.3 illustrates the architecture of the existing YOLOv11 model.

3.5.1 Backbone

The YOLO architecture, which is a pre-trained CNN that applies enormous datasets to extract the features from the input frame, significantly depends on the backbone. Feature maps are generated with the use of convolutional layers.

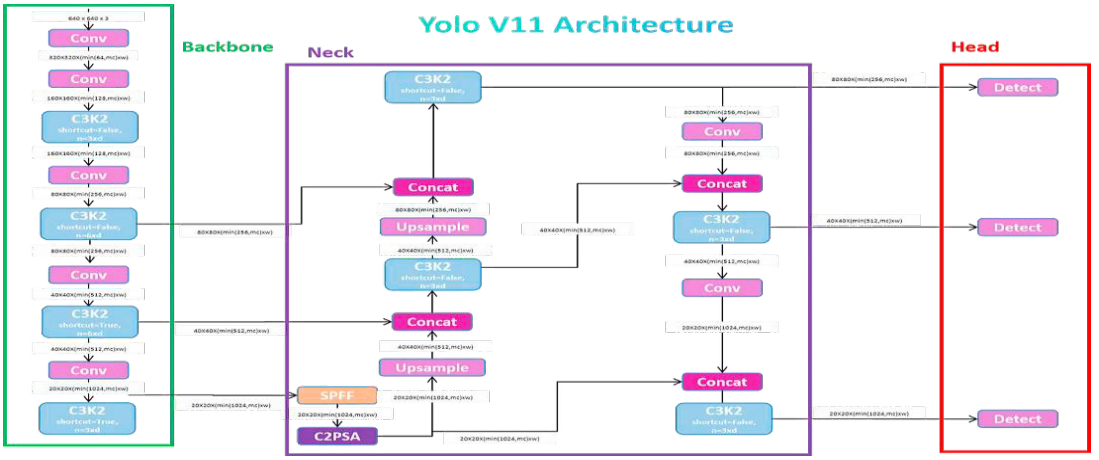


Fig. 3: The architecture of the YOLOv11 model

3.5.2 Convolutional Layers

The YOLOv11 model is assembled through a predecessor by utilising the convolutional layers for input frame down sampling, which enables the progressive reduction of. For enhanced representation, expand the number of feature channels while retaining spatial dimensions. The YOLOv11 model uses the C3k2 block during this feature extraction stage, which takes the place of the C2f block in the previous iteration. It combines two smaller convolutional techniques into one large one, the C3k2 block improves the Cross-Stage Partial (CSP) Bottleneck's computational efficiency and cost. A smaller kernel size in the input layer is represented by the 'k2' module in C3k2, which helps to achieve more rapid processing speeds without reducing detection performance.

3.5.3 SPPF and C2PSA

The Spatial Pyramid Pooling module (SPPF), which improves the design by using a new Cross Stage Partial with Spatial Attention (C2PSA) block, is still present in the YOLOv11 models and the classification of inner layers in YOLOv11 model is explained in Fig. 4. To further reinforce the spatial attention mechanism in the feature maps, the C2PSA block is also introduced. The C2PSA block enhances the detection accuracy, particularly for small potholes, and enables YOLOv11 to focus on critical areas by applying the spatial pooling layers to the information.

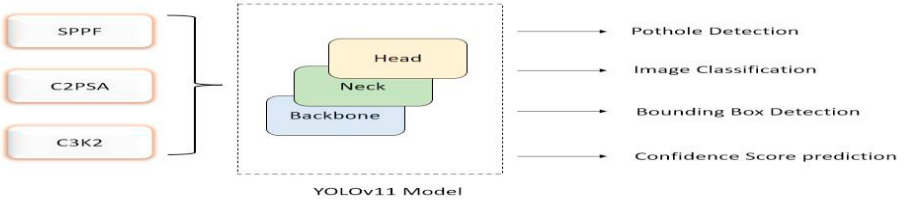


Fig. 4: Classification of Inner layers in the YOLOv11 model

3.5.4 Neck

In the neck module, the model used to predict the final class, in the YOLO architecture's neck module, combines the information from multiple scales and transmits them to the head module. In order to efficiently capture and use the multi-scale information, this approach usually uses an upsampling module that concatenates feature maps from different stages.

3.5.5 C3K2 Block

The C3k2 block enhances the overall performance of the feature aggregation method by enhancing detection speed and efficiency. The frame is handled by concatenation and up sampling, the neck module in YOLOv11 integrates to enhance the C3k2 block, which raises the model speed and performance.

3.5.6 Attention Mechanism

The C2PSA module in addition to YOLOv11 that includes C2PSA focuses on some specific regions of the frame due to its attention strategy which results in a more accurate detection, especially for small or partially visible potholes. In contrast to its predecessor, YOLOv8 lacks in this specific attention strategy.

3.5.7 Head

The final predictions for object identification and categorization are generated by the YOLOv11 head module. After analysing the feature maps transmitted from the neck, it generates bounding boxes and class designations for the objects in the frame.

3.5.8 CBS Block

CBS (Convolution-Batch Norm-Silu) layers are included in the head module of YOLOv11, following the processing of the C3k2 blocks. Using a feature map, these layers further improve the process and predict a more accurate detection result by using relevant features. A data flow that is both normalizing and stabilizing is used in the batch normalization process. The module enhances the performance of the model by using the SiLU activation function for non-linearity.

4. **Result and Discussion**

This section discusses the practical work used to achieve the results of the proposed work and depicts the relevant requirements that were used to experiment, such as the AMD Ryzen 5 5800H CPU. This work is experimented with a platform of a 64-bit operating system, 16 GB of RAM, and a 512 GB SSD for our test configuration. In the investigation, an NVIDIA RTX T4100 GPU operated on the computational tasks. In this section, Fig. 5 depicts the qualitative results of the enhanced YOLOv11-x and YOLOv8-x models.

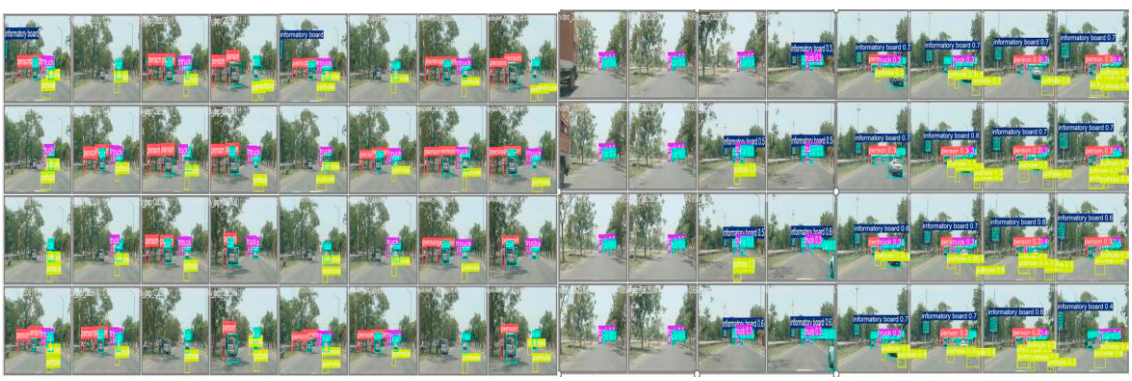


Fig. 5: Qualitative results of enhanced YOLOv11-x and YOLOv8-x models

The proposed results demonstrate better performance against considered models *i.e.* YOLOv8-x and YOLOv11. In YOLOv8-x, being a lightweight version, it demonstrated faster inference but compromised on detection quality. Table 2 represents the Quantitative results of both YOLO variants and non-YOLO models.

Table 2: Quantitative results of the YOLO and non-YOLO Models

Model	Precision	Recall	F1-score	mAP@50	mAP@50-95
YOLOv8-x (Improved)	1.00	0.65	0.41	0.477	0.22
YOLOv8-x	1.00	0.63	0.38	0.469	0.21
YOLOv11-x (Improved)	1.00	0.67	0.42	0.506	0.23
YOLOv11-x	1.00	0.67	0.39	0.449	0.21
SSD-MobileNet v2 [19]	0.71	0.56	0.62	0.49	0.20
Faster R-CNN (Resnet-50) [20]	0.84	0.60	0.69	0.58	0.21

Object detection models YOLOv8-x, YOLOv11-x and two non-YOLO models: SSD-MobileNet v2 & Faster R-CNN are compared in Table 2 using the following essential evaluation metrics: Precision, recall, F_1-score, mAP@50, and mAP@50-95. YOLOv8x models have lower recall values (0.65-0.63), meaning they miss a number of true detections, but they perform better in terms of accuracy, i.e., (1.00), which shows that there are no false positive predictions. Their mAP@50 and mAP@50-95 scores (0.477-0.469 and 0.22-0.21, respectively) are improved, suggesting moderate detection accuracy, while their F_1-scores are comparatively low (0.41-0.38), reflecting the better trade-off between precision and recall. In contrast, the YOLOv11-x proposed models show better recall (0.67) and slightly higher F_1-score (0.42). YOLOv11-x (proposed) models outperform on overall metrics, achieving the highest values, indicating superior overall detection performance. This implies that the YOLOv11-x proposed model shows a better balance between detecting more true objects and maintaining general accuracy, resulting in more effective model performance for object detection and image classification. The model uses a feature extraction technique and class discrimination capabilities, especially in complex or cluttered scenes. The background class also interferes with object predictions, due to insufficient boundary learning or annotation issues. These results highlight the importance of improving class separation through better data augmentation, fine-tuned architectures, or class-balanced training strategies. Normalised confusion matrices thus serve as a critical diagnostic tool to guide model refinement and ensure reliable multi-class classification in real-time scenarios. The performance qualities of a YOLOv8 object detection model over multiple epochs are shown in the training and validation loss in this image. However, the evaluation metrics, such as recall and precision plots, exhibit fluctuations rather than stable improvements, suggesting inconsistency in how confidently and completely the model identifies objects across epochs, and the inconsistency factors due to overfitting.

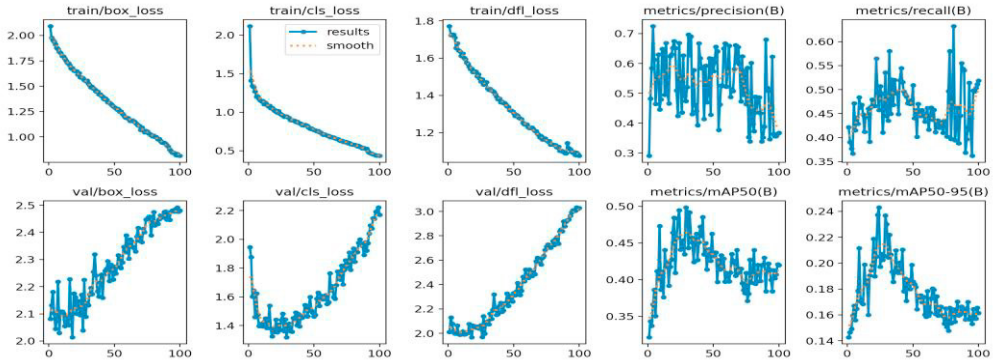


Fig. 6: Training and validation loss with mean average precision using YOLOv8-x (proposed)

In the bottom row, the validation losses (box, class, and DFL) initially decrease but then begin to rise, overfitting the model fits the training data well but starts performing worse on unseen data, as shown in Fig. 6. The training and validation curves of the YOLOv11x model over 50 epochs are shown in Fig. 7 which shows both loss functions and evaluation metrics. During training, the box, classification, and distribution focal losses exhibits a consistent downward

trend, indicating effective optimization and improved localization and classification performance.

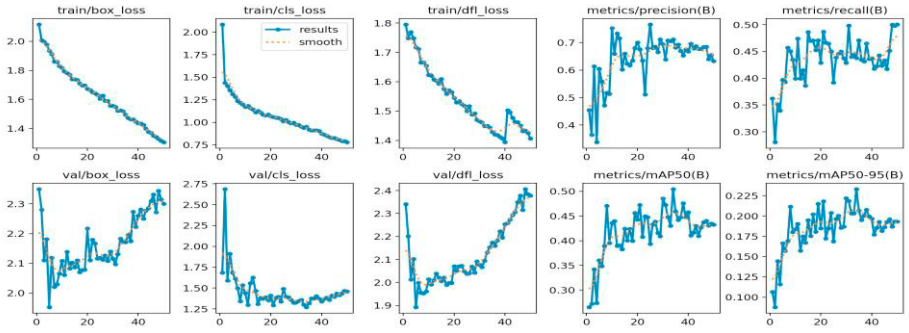


Fig. 7: Training and validation loss with mean average precision using YOLOv11-x (proposed)

Precision and recall gradually increases reflecting better detection capability over time but the validation losses initially decrease and begin to rise after around 20 epochs to fit the training data. Despite this, the mAP@50 and mAP@5095 metrics represent the continued enhancement, especially up to around epoch 30, indicating that the model's detection accuracy across various Intersection-over-Union thresholds is increasing.

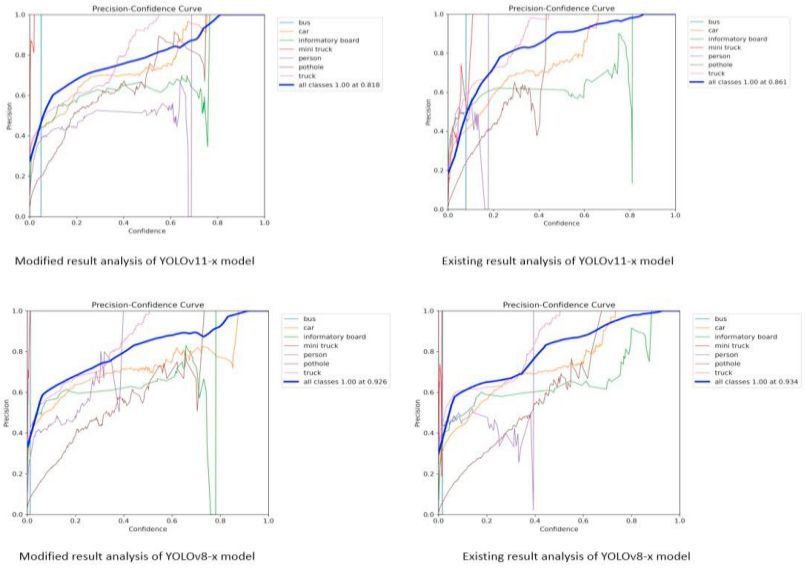


Fig. 8: Comparative analysis of the Precision- Confidence curves for the YOLOv8-x & YOLOv11-x Model

The Fig.8 illustrates a comparative analysis of the Precision-Confidence curves for the Proposed YOLOv11-x & YOLOv8-x model. In the proposed model, all classes show that precision peaks at 0.818, whereas in the existing model, it achieves a slightly higher value of 0.861. Individual class performance shows variation for categories such as "car" and "truck," while "pothole" and "mini truck" exhibit more changing behaviour, reflecting variability in detection consistency across classes. Both models show the increasing precision with higher confidence thresholds, but the comparison shows that the modified YOLOv11-x model demonstrates slightly reduced aggregate precision as compared to the existing version, but performance of the modified model can be enhanced if refinement and tuning of the dataset are done and the overall model performance is strong with all classes at higher confidence values, 0.926 for the modified results.

5. Conclusion

This paper proposed a model that uses a deep learning enabled model i.e. YOLOv11-x and YOLOv8-x for detecting potholes and image classification. The pothole-based videos have been captured from a dashcam device and converted into frames, and then annotated using roboflow for further processing. These frames are pre-processed with the auto-orient method and resized to 640 x 640. The model improves the evaluation of metrics through precision, recall and F_1-score. In this work, the improved YOLOv11-x and YOLOv8-x models have been compared with considered YOLO models and non-YOLO models (SSD-MobileNet, and Faster R-CNN). These are experimented over the same dataset using identical training and testing splits. The outcome of the proposed YOLO models significantly delivers outranking performance against considered YOLO and non-YOLO models. In future work, the proposed work will be extended to experiment on high performance GPUs specially in difficult dashcam scenarios such as low light, wet roads, shadows and motion blur to deliver better solution for ADAS system.

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